

SAN CARLOS DE BARILOCHE, Argentina

WMO: 877650

Lat: 41.1475S

Lon: 71.1642W

Elev: 2756

StdP: 13.29

Time Zone: -3.00 (W03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.6	22.8	12.2	11.3	35.5	15.5	13.3	35.7	39.4	44.1	34.2	42.4	3.4	140	0.785

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	26.7	84.0	58.2	80.3	56.5	76.8	55.3	60.1	78.2	58.2	75.6	56.6	72.9	11.6	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
53.2	66.6	64.3	50.9	61.1	62.2	49.2	57.4	60.7	27.7	78.2	26.3	75.3	25.3	72.9	70.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
34.0	30.7	28.0	11.9	90.1	5.8	2.6	7.8	92.0	4.4	93.5	1.1	94.9	-3.1	96.8
			10.9	64.1	5.7	2.2	6.8	65.7	3.5	67.1	0.3	68.3	-3.8	69.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	47.6	59.2	59.0	54.1	47.5	42.4	38.0	36.7	38.5	42.2	46.8	51.2	55.8	(6)
	DBStd	9.48	5.71	6.13	5.57	5.27	5.15	5.28	5.31	4.74	4.79	4.96	5.30	5.79	(7)
	HDD50	1933	4	4	21	108	240	360	413	357	238	127	46	15	(8)
	HDD65	6406	195	185	340	524	700	810	878	821	685	565	414	289	(9)
	CDD50	1042	289	256	149	34	5	1	0	1	3	26	82	196	(10)
	CDD65	40	15	17	3	0	0	0	0	0	0	0	0	5	(11)
	CDH74	1343	478	455	119	3	0	0	0	0	0	8	52	228	(12)
	CDH80	326	125	129	21	0	0	0	0	0	0	1	5	45	(13)
(14) Wind	WSAvg	12.6	14.9	12.7	12.0	11.4	9.8	11.1	10.9	11.3	11.9	13.9	15.8	16.1	(14)
(15) Precipitation	PrecAvg	31.1	0.9	0.8	1.1	2.3	4.8	6.0	4.9	4.5	2.2	1.7	1.1	1.1	(15)
	PrecMax	44.8	4.2	3.3	5.1	6.9	13.5	15.7	9.5	12.1	6.5	7.2	4.4	7.5	(16)
	PrecMin	19.9	0.0	0.0	0.0	0.0	0.3	0.2	0.6	0.6	0.1	0.0	0.0	0.0	(17)
	PrecStd	6.3	1.0	0.8	1.0	1.7	3.3	3.4	2.2	2.6	1.5	1.6	1.0	1.3	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	88.1	89.1	82.6	72.0	60.9	54.3	53.7	58.8	65.8	73.6	80.2	84.6	(19)
		MCWB	59.4	60.2	59.2	53.4	49.6	46.0	44.5	46.2	47.6	50.8	54.4	58.3	(20)
	2%	DB	83.8	84.1	77.1	67.1	56.9	50.4	49.9	53.3	60.7	67.9	74.0	80.1	(21)
		MCWB	58.0	58.1	56.3	51.3	48.2	44.5	43.2	44.3	45.4	48.5	52.3	55.6	(22)
	5%	DB	79.5	79.7	72.5	62.5	53.8	48.3	47.0	49.9	56.8	63.0	69.7	75.5	(23)
		MCWB	56.3	56.8	54.6	49.4	46.8	43.0	41.6	42.5	43.8	47.0	50.5	54.1	(24)
	10%	DB	75.3	75.3	68.2	58.7	51.5	46.2	44.8	46.8	52.8	58.9	64.8	71.3	(25)
		MCWB	54.8	55.2	52.7	48.6	45.5	41.8	40.2	41.1	42.4	45.8	48.7	52.7	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	63.0	62.2	61.4	55.6	52.2	48.3	47.5	48.3	49.3	53.6	55.7	60.0	(27)
		MCDB	81.1	81.6	78.6	67.0	57.3	52.1	50.4	54.6	61.9	67.0	75.2	79.7	(28)
	2%	WB	59.9	60.1	58.0	53.2	50.0	45.9	44.3	45.9	47.0	50.6	53.6	57.4	(29)
		MCDB	77.6	78.7	73.2	62.7	55.4	49.7	48.0	51.1	57.5	64.0	70.3	76.4	(30)
	5%	WB	57.9	58.3	55.8	51.3	47.8	43.8	42.3	43.7	45.0	48.5	51.8	55.4	(31)
		MCDB	75.6	75.4	69.4	59.6	52.6	47.4	46.1	48.7	53.9	60.2	66.7	72.2	(32)
	10%	WB	56.1	56.5	53.8	49.5	46.2	41.9	40.5	41.4	43.3	46.8	49.9	53.5	(33)
		MCDB	72.2	72.2	65.7	56.9	50.5	45.3	44.1	45.8	51.0	57.1	62.7	68.8	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	26.7	27.3	25.4	21.2	15.8	12.7	14.0	15.2	19.4	21.3	23.2	24.4	(35)
		MCDBR	37.2	37.2	35.2	31.4	20.7	16.0	18.3	21.2	28.7	32.6	34.8	36.4	(36)
	5% WB	MCWBR	16.5	16.8	18.0	18.2	14.0	11.8	13.0	13.7	16.3	17.0	17.0	16.7	(37)
		MCWBR	32.2	32.3	30.2	24.7	17.3	13.2	15.2	17.5	23.4	26.7	29.2	31.5	(38)
(40) Clear-Sky Solar Irradiance	taub		0.298	0.304	0.294	0.289	0.281	0.271	0.267	0.284	0.296	0.298	0.298	0.302	(40)
		taud	2.592	2.567	2.595	2.592	2.589	2.555	2.552	2.522	2.514	2.537	2.560	2.558	(41)
	Ebn at Noon		327	317	307	287	268	260	271	284	301	316	325	328	(42)
		Edh at Noon	32	31	28	24	20	19	21	25	30	32	33	34	(43)
(44) All-Sky Solar Radiation	RadAvg		2509	2166	1607	1042	623	454	540	766	1230	1694	2202	2478	(44)
	RadStd		119	105	91	84	84	65	62	82	134	128	143	183	(45)

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	+1.55	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (2 neighbors)	N/A	N/A	N/A	+1.01	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page